UNIVERSITY OF CAPE TOWN

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II Advanced Calculus (2AC)

Functions of several variables: Limits - Continuity

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0.1 The domain

What is the domain of a function of two variables and how to find such a domain?

In general, this depends on the way the function is given or presented.

1. If a function is given as $f(x, y) :=$ ..... with other information, then its domains is the set D of all points (x, y) in $\mathbb{R}^2$ except those points at which $f(x, y)$ cannot be defined. There are only two situations in which a given function cannot be defined at a point (a, b):

$$\boxed{\text{either the value of } f(a,b) \text{ is not a real number or } f(a,b) \text{ takes values } \pm\infty}. \qquad (*)$$

For instance,

$$\boxed{f_1(x, y) := \sqrt{x^2 + y^2 - 1} \quad \text{or} \quad f_2(x, y) = \frac{1}{x^2 + y^2}}. \tag{1}$$

For points (a, b) such that $a^2 + b^2 < 1$, we get $a^2 + b^2 - 1 < 0$ and therefore, in that case $\sqrt{a^2 + b^2 - 1}$ will be a purely imaginary complex number (hence, non real!). So, the function f_1 is not defined at such points (a, b).

For the second function f_2, notice that for $(a, b) = (0, 0)$, "the value of $f_2(a, b)$ is ∞."

This gives us a practical approach to find the domain of a given function f which consists to first find the set of those points at which the function is not defined and hence, to deduce the domain as the complement of that set. For instance, to find the domain of the function

$$\boxed{f(x, y) := \frac{2021}{16 - x^2 - y^2}}$$

we find all the points for which the denominator is equal to 0, that is, the circle $x^2 + y^2 = 16$. Therefore, the domain is the set of the points (x, y) outside that circle.

Notice that the domain of the function f_2 in (1) is $\mathbb{R}^2 \setminus \{(0, 0)\}$.

2. If the function is given in the form $f : E \subset \mathbb{R}^2 \to \mathbb{R}$, one has to take into account the set $E \subset \mathbb{R}^2$. Therefore, the domain of the function f in this case is the set of points (x, y) in the set E such that the function f is defined as specfied in (*). For instance, the function $f : (-3, 4) \times (0, 3) \to \mathbb{R}$ defined by

$$f(x, y) := 1 - \ln(|x| + (y - 2)^2)$$

has domain

$$D_f := \big\{(x, y) \in (-3, 4) \times (0, 3)\colon\ (x, y) \neq (0, 2)\big\}\,.$$

Exercise 0.1 Describe **or** sketch the domain of each function below:

(a) $f_1(x, y) := \dfrac{2021x - 2020}{1 - \ln(2x^2 + y^2)}$;

(b) $f_2(x, y) := \sqrt{\tan(xy)}\,\cos\big(e^{\sqrt{xy}}\big)$;

(c) $f_3(x, y) := \sqrt{\cos(x^2 + y^2)}$;

(d) $f_4(x, y) := (\sin x)^y + (\sin y)^x$ for $x, y > 0$;

(e) $f_5(x, y) := \dfrac{x^2 + \sin^2 y}{2x}\tan\big(e^{x+y}\big)$.

0.2 Level sets and graphs

Definition 0.2 (Graph)
Let $f : E \subset \mathbb{R}^n \to \mathbb{R}$. The graph of the function f is the set

$$\boxed{G_f := \big\{(x_1, x_2, \dots, x_n, f(x_1, x_2, \dots, x_n))\colon\ (x_1, x_2, \dots, x_n) \in E\big\}}\,.$$

Definition 0.3 (Level sets)
Let $f : E \subset \mathbb{R}^n \to \mathbb{R}$. The level sets of the function f are the sets

$$\boxed{L_c := \big\{(x_1, x_2, \dots, x_n) \in E\colon\ f(x_1, x_2, \dots, x_n) = c\big\},\quad c \in \mathbb{R} \text{ varying}}\,.$$

- For $n = 2$ we talk about *level curves.*
- For $n = 3$ we talk about *level surfaces.*

Example 0.4
Consider the function $f(x, y) = x^2 + y^2$ for which the level curves with $c \geq 0$ are circles centered at $(0, 0)$ with radius $\sqrt{c}$, while the graph of the function f in this case is called a *paraboloid.*

For the function $f(x, y) := \sqrt{x^2 + y^2}$ the level curves for $c \geq 0$ are the circles centered at $(0, 0)$ with radius c and the graph is called a *circular cone.*

The function $f(x, y) := \sqrt{2x^2 + 3y^2}$ has level curves for $c \geq 0$, the ellipses $2x^2 + 3y^2 = c$ and the graph is called *spherical cone.*

Exercise 0.5
Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$\boxed{f(x, y) := \frac{2x - 2y}{x^2 + y^2 + 2}\quad \forall (x, y) \in \mathbb{R}^2}\,.$$

(a) Determine the level sets $\{f = k\} = \{(x, y) \in \mathbb{R}^2\colon\ f(x, y) = k\}$ of the function f.

(b) Determine the level set on which lies the point $(0, -1)$.

(c) Sketch the level sets $\{f = \sqrt{2}/2\}$, $\{f = 0\}$ and $\{f = -\sqrt{2}/2\}$.

Answer:(a) $f(x, y) = k \ \Leftrightarrow\ \dfrac{2x - 2y}{x^2 + y^2 + 2} = k \ \Leftrightarrow\ x^2 + y^2 + 2 = \dfrac{2}{k}x - \dfrac{2}{k}y$ if $k \neq 0$ and complete the squares $x^2 - \dfrac{2}{k}x = \ldots$ and $y^2 + \dfrac{2}{k}y = \ldots.$
(b) The level set is the set of points (x, y) such that $f(x, y) = f(0, -1)$ and find the level set.
(c) Sketch the level sets. For instance, $f(x, y) = \sqrt{2}/2 \ \Leftrightarrow\ \dfrac{2x - 2y}{x^2 + y^2 + 2} = k$
$\Leftrightarrow\ x^2 + y^2 + 2 = 2\sqrt{2}x - 2\sqrt{2}y \ \Leftrightarrow\ (x - \sqrt{2})^2 + (y + \sqrt{2})^2 = 2$, circle of centre $(\sqrt{2}, -\sqrt{2})$ and radius $\sqrt{2}$. Find the other two level curves and sketch them.

0.3 Limits and continuity

To illustrate the notion of limit, we will consider in this section functions $f : \mathbb{R}^2 \to \mathbb{R}$.
Consider the function $f(x, y) = 3x - y$ and let us give to (x, y) values which are getting closer to the point $(1, 1)$ and see what happens to the corresponding numbers $(f(x, y)$.
We will notice that as (x, y) becomes to the point $(1, 1)$, the numbers $f(x, y)$ are getting closer the number 2. Therefore, we say that the function f has limit at $(1, 1)$ equal to the number 2 and we write

$$\boxed{\lim_{(x,y)\to(1,1)} f(x, y) = 2}.$$

Definition 0.6 (Limits)
We say that the function $f : \mathbb{R}^2 \to \mathbb{R}$ has limit equal L when (x, y) approaches the point (a, b) (written $\lim_{(x,y)\to(a,b)} f(x, y) = L$) if the numbers $f(x, y)$ are made **arbitrarily** close to the number L by taking (x, y) **sufficiently** close the point (a, b).

One could also reformulate $\lim_{(x,y)\to(a,b)} f(x, y) = L \in \mathbb{R}$ as follows: $f(x, y)$ approaches L **indefinitely** as (x, y) approches (a, b) **independently** of the direction in which (x, y) approches (a, b).

Using mathematical symbols, this can be illustrated as follows: For every $\varepsilon > 0$, if we consider the open interval $(L - \varepsilon, L + \varepsilon)$ we can always find a small disc $D((a, b), \delta)$ centered at the point (a, b) with radius $\delta > 0$ such that the image $f(D((a, b), \delta)) \subset (L - \varepsilon, L + \varepsilon)$ i.e.,

$$\boxed{\text{For every } (x, y) \in D((a, b), \delta), \quad \text{it follows that} \quad f(x, y) \in (L - \varepsilon, L + \varepsilon)}.$$

Remark 0.7 In a situation when the domain of the function is not the whole $\mathbb{R}^2$ (as it happens in many cases), i.e., $f : E \subsetneq \mathbb{R}^2 \to \mathbb{R}$, then in order to talk about the limit $\lim_{(x,y)\to(a,b)} f(x, y)$, the given point (a, b) should be such that every disc centered at (a, b) should contain points of the domain E which are different from the point (a, b). That is, for every $\delta > 0$ we should have that

$$D((a, b), \delta) \cap (E \setminus \{(a, b)\}) \neq \emptyset\,.$$

In that case, we say that (a, b) is *an accumulation point* for the domain E.

In general, two distinguished cases arise:

0.3.1 The limit exists

In this case one should prove the existence of the limit. Either one substitute (a, b) into the expression of the function in that case we will talk about continity of f at (a, b) (to be studied later) or some estimates which might lead to the application of the squeeze theorem below.

Theorem 0.8 (The squeeze theorem)
Let the functions $f, g, h : \mathbb{R}^2 \to \mathbb{R}$ be such that

$$\boxed{g(x, y) \leq f(x, y) \leq h(x, y) \qquad \textit{for } (x, y) \textit{ in a disc } D \textit{ centered at } (a, b)}.$$

If

$$\boxed{\lim_{(x,y)\to(a,b)} g(x, y) = \lim_{(x,y)\to(a,b)} h(x, y) = L, \ \textit{then} \ \lim_{(x,y)\to(a,b)} f(x, y) = L}.$$

For instance, to find the limit $\displaystyle\lim_{(x,y)\to(0,0)} \frac{xy^2}{x^2 + y^2}$ one could notice that

$$0 \leq \left| \frac{xy^2}{x^2 + y^2} \right| = |x| \left(\frac{y^2}{x^2 + y^2} \right) \leq |x|$$

and then apply the squeeze theorem to find that the limit is equal to zero.

Quite often, the existence of the limit is obtained through the use of polar coodinates r, θ with

$$\boxed{x = a + r\cos\theta \qquad \text{and} \qquad y = b + r\sin\theta}.$$

To simplify our presentation we formulate the following important result on finding the limit through the polar coordinates r, θ in the case where $(a, b) = (0, 0)$.

Theorem 0.9 (A criterion for limit through polar coordinates)
In order to have $\displaystyle\lim_{(x,y)\to(0,0)} f(x, y) = L$ it is sufficient to write an estimate of the type

$$\boxed{\big|f(r\cos\theta, r\sin\theta) - L\big| \leq g(r) \qquad \textit{with} \qquad \lim_{r\to 0^+} g(r) = 0}. \tag{2}$$

It is essential to have a function g which does not depend on θ. In this case, we say that

$$\boxed{\lim_{r\to 0^+} f(r\cos\theta, r\sin\theta) = L \qquad \textit{uniformly in } \theta}.$$

Practically, one first needs to identity the number L. It is enough to find along the axes in order to identify the candidate L to be used in (2) above and then find a function g which will satisfy (2). The function g is often determined through our knowledge of the trig. identities. For example, find the limit

$$\lim_{(x,y)\to(0,0)} \frac{y^2 \ln(1 + x)}{x^2 + y^2}.$$

we first notice that the limit is equal to 0 along both axes: the x-axis ($y = 0$) and the y-axis ($x = 0$). Therefore, the candidate as limit is the number $L = 0$. Now, we write

$$f(x, y) = f(r\cos\theta, r\sin\theta) = \frac{r^2\sin^2\theta\ln(1 + r\cos\theta)}{r^2} = \sin^2\theta\ln(1 + r\cos\theta)\,.$$

Now, using the fact that $\lim_{t\to 0}\frac{\ln(1+t)}{t} = 1$ we get that $\ln(1+t) \approx t$ for $|t|$ small. So, $\ln(1+r\cos\theta) \approx r\cos\theta$ for r small enough. Therefore, $\sin^2\theta\ln(1+r\cos\theta) \approx r\sin^2\theta\cos\theta$ for r small enough. Thus,

$$0 \le |f(r\cos\theta, r\sin\theta)| \approx r\sin^2\theta|\cos\theta| \le r$$

and hence, $\lim_{r\to 0} f(r\cos\theta, r\sin\theta) = 0$ uniformly in θ.

One could also find the limit directly by noticing that $\frac{y^2}{x^2+y^2} \le 1$ and hence,

$$0 \le \left|\frac{y^2\ln(1+x)}{x^2+y^2}\right| \le |\ln(1+x)| \to 0 \qquad \text{as } (x, y) \to (0, 0)$$

and apply the squeeze theorem to find that $\lim_{(x,y)\to(0,0)}\frac{y^2\ln(1+x)}{x^2+y^2} = 0\,.$

Remark 0.10 The following identities and inequalities are often used when finding the limits through the estimate method:

$$\boxed{\cos^2 t + \sin^2 t = 1 \;\Rightarrow\; \begin{cases} |\sin t| \le 1; \\ |\cos t| \le 1; \end{cases} \quad \text{and also} \quad \begin{cases} |\sin t| \le |t|; \\ \ln(1+t) \le t; \\ e^t \ge t + 1; \end{cases} \quad \begin{cases} |\alpha \pm \beta| \le |\alpha| + \beta| \\ \alpha\beta \le \frac{1}{2}(\alpha^2 + \beta^2); \\ \frac{\alpha^2}{\alpha^2+\beta^2} \le 1\,. \end{cases}}$$

Definition 0.11 (Continuity)
The function $f : E \subset \mathbb{R}^2 \to \mathbb{R}$ is said to be continuous at (a, b) if

$$\boxed{\lim_{(x,y)\to(a,b)} f(x, y) = f(a, b)}\,.$$

Remark 0.12 The definition of continuity is equivalent to the following three requirements:

(i) the function f is defined at the point (a, b);

(ii) the limits exists;

(iii) the limit is equal to $f(a, b)$.

Whenever any of the three conditions (i)-(iii) fails, we say that the function f is *discontinuous* at the point (a, b) and failure of each of these conditions gives rise to a different type of discontinuity.

0.3.2 The limit does not exist

We mentioned above that the existence of the limit $\lim_{(x,y)\to(a,b)} f(x,y) = L$ is independent of the path that the running points (x,y) follow when approachng the point (a,b). Therefore, in order to show that the limit does not exist it is sufficient to find two curves $\mathcal{C}_1$ and $\mathcal{C}_2$ passing through the point (a,b) along which we get two different limits. For example consider the function $f(x,y) := \dfrac{x^2y}{x^4+y^2}$. One can notice that it has limit equal to 0 along both axes while along the curve $y = x^2$ we find that $f(x,x^2) = \dfrac{x^4}{2x^4} = \dfrac{1}{2}$ and hence, the limit along that curve is equal to $\dfrac{1}{2} \neq 0$. Therefore, the limit does not exist.

0.3.3 The infinite limit

Definition 0.13 (Infinite limit)
(i) We say that the function $f : \mathbb{R}^2 \to \mathbb{R}$ has limit equal to $-\infty$ at the point (a,b) if the number $f(x,y)$ can be made smaller that any fixed negative number by choosing the points (x,y) sufficiently close to the point (a,b).
Using mathematical symbols, this can be expressed as follows: for any number $M < 0$, we can always find a small disc $D((a,b),\delta)$ centered at the point (a,b) with radius $\delta > 0$ such that the image $f(D((a,b),\delta)) \subset (-\infty, M)$ i.e.,

$$\boxed{\text{for every } (x,y) \in D((a,b),\delta) \quad \text{it follows that} \quad f(x,y) < M}.$$

(ii) Similarly, we say that the function $f : \mathbb{R}^2 \to \mathbb{R}$ has limit equal to ∞ at the point (a,b) if the number $f(x,y)$ can be made greater than any fixed positive number by choosing the points (x,y) sufficiently close to the point (a,b).
Using mathematical symbols, this can be expressed as follows: for any number $M > 0$, we can always find a small disc $D((a,b),\delta)$ centered at the point (a,b) with radius $\delta > 0$ such that the image $f(D((a,b),\delta)) \subset (M, \infty)$ i.e.,

$$\boxed{\text{for every } (x,y) \in D((a,b),\delta) \quad \text{it follows that} \quad f(x,y) > M}.$$

0.3.4 Some exercises on various limits

Exercise 0.14
Find the following limits (if they exists) $\lim_{(x,y)\to(0,0)} f_i(x,y)$ for $i = 1, \dots 12$:

(a) $f_1(x,y) := \dfrac{xy^2 - x^2}{x^2 - 2xy + y^2}$; $\boxed{\text{Ans: } \nexists}$;

(b) $f_2(x,y) := \dfrac{x^2 + \sin^2 y}{2y} e^{x+y}$; $\boxed{\text{Ans: } \nexists}$;

(c) $f_3(x,y) := \dfrac{x - y}{x + \sin y}$; $\boxed{\text{Ans: } \nexists}$;

(d) $f_4(x,y) := \dfrac{\arctan(x^2 + y^2)}{1 + e^{-x-y}}$; $\boxed{\text{Ans: } 0}$;

(e) $f_5(x,y) := x^2 y \ln(x^4 + y^2) - \pi$; $\boxed{\text{Ans: } -\pi}$;

(f) $f_6(x,y) := \sin(x) \ln y$; $\boxed{\text{Ans: } \nexists}$;

(g) $f_7(x,y) := \dfrac{x^3 y + 2\sin(xy^2)\cos(2x + y)}{x^2 + y^2}$; $\boxed{\text{Ans: } 0}$;

(h) $f_8(x,y) := \dfrac{xy}{x^2 - xy + y^2}$; $\boxed{\text{Ans: } \nexists}$;

(i) $f_9(x,y) := \dfrac{\sin x}{\sqrt{x^2 + y^2}}$; $\boxed{\text{Ans: } \nexists}$;

(j) $f_{10}(x,y) := (\cos x - 1) \ln y$; $\boxed{\text{Ans: } \nexists}$;

(k) $f_{11}(x,y) := \dfrac{x^{\frac{1}{4}} y^{\frac{7}{3}}}{x^2 + y^2}$; $\boxed{\text{Ans: } 0}$;

(l) $f_{12}(x,y) := \dfrac{\cos(2x) - 1}{\sqrt{x^2 + y^2}}$; $\boxed{\text{Ans: } 0}$;

(m) $f_{13}(x,y) := \dfrac{\sin(x^3 + y^3)}{x^2 + y^2}$; $\boxed{\text{Ans: } 0}$;

(n) $f_{14}(x,y) := \dfrac{|\sin(y)| x}{x^2 + \sqrt{|y|}}$; $\boxed{\text{Ans: } 0}$.

Remark 0.15 (Limit at infinity)
If the domain E of the function f is unbounded, one may also be interested in finding the limit

$$\lim_{x^2 + y^2 \to \infty} f(x,y)$$

for which one may have all the cases studied above. For instance, the case $\lim\limits_{x^2 + y^2 \to \infty} f(x,y) = L$ means that $f(x,y)$ are made **arbitrarily** close to the number L by taking the points (x,y) outside a certain disc centered at the origin.
Using mathematical symbols, this can be expressed as follows: for any number $\varepsilon > 0$, we can always find a disc $D((0,0),\delta)$ centered at the point $(0,0)$ with radius $\delta > 0$ outof which the function f takes values in the interval $(L - \varepsilon, L + \varepsilon)$. i.e.,

$$\boxed{\text{for every } (x,y) \in E, \quad (x,y) \notin D((a,b),\delta) \quad \Rightarrow \quad f(x,y) \in (L - \varepsilon, L + \varepsilon)}.$$

We could expand this further but the syllabus of this course is restricted only to the notion of limit at a point!

Exercise 0.16 Study the continuity of the function at the given point:

(a) $f_1(x,y) := \begin{cases} y \ln(x^2 + y^2) & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$ at $(0,0)$; $\boxed{\text{Ans: continuous}}$;

(b) $f_2(x,y) := \begin{cases} \dfrac{x^{\frac{5}{3}} y^{\frac{2}{5}}}{x^2 + y^2} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$ at $(0,0)$; $\boxed{\text{Ans: continuous}}$;

(c) $f_3(x,y) := \begin{cases} 1 & \text{if } y > x^2, \\ 0 & \text{if } y \leq x^2. \end{cases}$ at $(0,0)$; $\boxed{\text{Ans: not continuous}}$;

Exercise 0.17 Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x, y) := \begin{cases} \dfrac{y}{x^2} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0 \end{cases}$

(a) Study the continuity of the function f at the point $(0, 0)$.

(b) Study the continuity of the f at the point $(0, 0)$ as a functions defined on the domain $D := \{(x, y) \in \mathbb{R}^2\colon\ |y| \leq x \leq 1\}$.

Hint: Here you have been asked to check whether $\displaystyle\lim_{\substack{(x,y) \to (0,0) \\ (x,y) \in D}} f(x, y) = f(0, 0)$.

We will start the notion of differentiability in the next set of lecture notes!